Statement sheet: does the frozen Lean statement say what Balaban's printed lemma says?

Object. The Lean proposition `AdalyFormal.StatementLaneG.P428Target8` against the sentence on p. 428 of Balaban, *Propagators for lattice gauge theories in a background field*, CMP 99 (1985) (CMP99b below): "a positive definite operator $C^*\Delta_k C$ with a lower bound $\gamma_0 > 0$ independent of k and $U \dots$ for an arbitrary configuration U satisfying (3.35), (3.36) with $M\alpha_0$ sufficiently small". This sheet is meant to be read alone. Part A has one block per clause (print, Lean, plain words, deviation, a yes/no question). Part B writes the whole statement in ordinary notation with the quantifier order. Part C lists the readings a reader might reject, what changes, and which proved variant covers the alternative.

Conventions. *Print locations.* "p. 396, l. 6–10" counts text lines of the printed page from the first line under the running head, displays not counted; for other pages `pN:lM` gives the page N and a line number M of a page transcription (not included here); quotes are verbatim, mathematics in TEX. *Lean locations* are of the excerpt; full paths are in the appendix. *Deviation ids* (D1–D12, N1–N6, $\sqrt{2}$) are short labels for the deviations listed in each row. In Lean, a field of "extent" L_μ means $L_\mu + 1$ sites; ℓ is Balaban's L.

Part A. Clause by clause

A1 Lattice, torus, scale, distance

Print. CMP96 p.223 (`CMP96_p223:162`): "a distance between two points $x, y \in \eta Z^d$, $\eta > 0$, is given by the l^1-norm of the vector $x - y$ and is denoted by $|x - y|$, i.e. $|x - y| = \sum_{\mu=1}^{d} |x_\mu - y_\mu|$". Domains: $\Omega_j \subset T_\eta$ (p.393, CMP96 p.224 (2.1)). Unit lattice: $\Lambda \subset \Lambda_k = \Omega_k^{(k)}$ (p.427, `p427:139`).

The print does not say the lattice is periodic, nor its dimension in this clause, nor the value of η.

ZStatementP428.lean:42–43,58–59,66–67,286–287

```
def scaleUp (ell k : ℕ) (Lu : PeriodicLatticeExtent) : PeriodicLatticeExtent :=
  fun μ => (Lu μ + 1) * ell ^ k - 1
def pcd (N a b : ℕ) : ℕ :=
  min (if a ≤ b then b - a else a - b) (N - (if a ≤ b then b - a else a - b))
def pdist {L : PeriodicLatticeExtent} (x y : PeriodicLatticeSite L) : ℕ :=
  ∑ μ : Fin 3, pcd (L μ + 1) (x μ).val (y μ).val
def TorusOK (ell M : ℕ) (Lu : PeriodicLatticeExtent) : Prop :=
  (∀ μ : Fin 3, ell ∣ Lu μ + 1) ∧ (∀ μ : Fin 3, ell < Lu μ + 1) ∧ (∀ μ : Fin 3, M ∣ Lu μ + 1)
```

In words. The unit torus has $\mathrm{Lu}\,\mu + 1$ sites per direction; the fine torus (carrying U and the tower) has ℓ^k times as many, so the fine spacing is $\eta = \ell^{-k}$ (it appears below as `((ell:ℝ)^k)⁻¹`). Distance is the sum over the three axes of the shorter way round the circle. `TorusOK`: ℓ and M divide every unit side, and $\ell <$ side.

Deviation. Dimension 3, group SU(2), periodic torus, $\eta = \ell^{-k}$ and the wrap-around in the distance are conventions not printed. `TorusOK`: row D7 (fewer tori; blocking hygiene).

Check (yes/no). Is it acceptable, for this lemma, to work on a 3-torus with $\eta = \ell^{-k}$ and periodic l^1 distance, for tori whose sides are multiples of ℓ and M?

A2 The tower $\Omega_0 \supset \cdots \supset \Omega_k$ (`Admissible`)

Print. CMP96 p.224 (`CMP96_p224:138-54`): "We consider a sequence of domains $\Omega_1 \supset \Omega_2 \supset \dots \supset \Omega_k$, $\Omega_j \subset T_\eta \dots$ $\Omega_j = B^j(\Omega_j^{(j)})$, $\Omega_j^{(j)} \subset T^{(j)}_{L^j\eta}$ and it is a sum of big blocks, $(L^j\eta)^{-1}\mathrm{dist}(\Omega_j^c, \Omega_{j+1}) > RM$, M is a size of big blocks and R is a big positive integer which will be fixed later." CMP99b p.393 (`p393:122-40`): "the sequence (2.1) starts with Ω_0, thus we have $\Omega_0 \supset \Omega_1 \supset \cdots \supset \Omega_k \dots$ $\Omega_{k+1} = \varnothing \dots$ The condition (2.2) is unchanged." Big blocks: p.408 (`p408:129`): "elementary cubes (cells) of the lattice $ML^j\eta Z^d$".

ZStatementP428.lean:51–51,54–55,73–82

```
def bigSide (M ell j : ℕ) : ℕ := M * ell ^ j
def BlockClosed {L : PeriodicLatticeExtent} (s : ℕ) (S : Set (PeriodicLatticeSite L)) : Prop :=
  ∀ x ∈ S, ∀ y : PeriodicLatticeSite L, (∀ μ : Fin 3, (y μ).val / s = (x μ).val / s) → y ∈ S
def Admissible {L : PeriodicLatticeExtent} (ell M R k : ℕ) (S : Tower L) : Prop :=
  (∀ j : ℕ, S.Om (j + 1) ⊆ S.Om j) ∧
  (∀ j : ℕ, k < j → S.Om j = ∅) ∧
  (∀ j : ℕ, j ≤ k → ∀ μ : Fin 3, bigSide M ell j ∣ (L μ + 1)) ∧
  (∀ j : ℕ, j ≤ k → BlockClosed (bigSide M ell j) (S.Om j)) ∧
  (∀ j : ℕ, j ≤ k → ∀ x : PeriodicLatticeSite L, x ∉ S.Om j →
      ∀ y ∈ S.Om (j + 1), R * bigSide M ell j < pdist x y) ∧
  (∀ x : PeriodicLatticeSite L, x ∈ S.Om 0 ↔
      ∃ y ∈ S.Om 1, ∃ z : PeriodicLatticeSite L,
        (∀ μ : Fin 3, (z μ).val / M = (x μ).val / M) ∧ pdist z y ≤ R * M)
```

In words. Clauses in order: (1) nested; (2) empty above level k; (3) a level-j block (side $M\ell^j$ fine sites) tiles each side; (4) each Ω_j is a union of such blocks; (5) a point outside Ω_j is at distance $> RM\ell^j$ (fine sites; equivalently $> RM$ in units of ℓ^j) from Ω_{j+1}; (6) Ω_0 is the union of the M-blocks having a site within RM of Ω_1.

Deviation. Clause (6) is the p.396 choice of Ω_0 (A3). Clauses (3), tiling and padding above k are not printed (N4); ℓ^1 periodic distance (N4); block side read as $M\ell^j$ fine sites (p.408). Row N3/N4: identical to the printed conditions under reading A of T_1 (Lean `RBridge.admissible_iff_print`); see Part C.

Check (yes/no). (a) Are big blocks at level j cubes of side $M\ell^j$ fine sites? (b) Is the collar of (2.2) strict ($>$) in fine sites $R \cdot M\ell^j$?

A3 The innermost domain Ω_0 and Λ_0 (last clause of `Admissible`)

Print. CMP99b p.396, l.10–12: "The set Λ_0 in the above condition means $\Lambda_0 = \Omega_0 \backslash \Omega_1$, where $\Omega_0 = \{$a union of big blocks of the lattice T_1, such that their distances to Ω_1 are $\leq RM\}$." p.394 (`p394:174-77`): "Let us introduce a domain Ω_0 such that $\Omega_1 \subset \Omega_0$ and Ω_0 is a union of big blocks of the lattice T_1, e.g. we may take $\Omega_0 = \{$a union of big blocks in T_1, with distances to $\Omega_1 \leqq RM\}$, or we may add to Ω_1 a thinner layer of the big blocks surrounding Ω_1."

In words. Lean (the last conjunct of the excerpt in A2): Ω_0 is exactly the set of M-blocks at distance $\leq RM$ from Ω_1; $\Lambda_0 = \Omega_0 \setminus \Omega_1$ is the stratum of index 0 (A5).

Deviation. Two readings are made: T_1 is the fine lattice in its own unit spacing ("reading A"; the alternative "reading B" takes T_1 as the level-k unit lattice); and p.394 allows a thinner layer, while Lean fixes the $\leq RM$ layer. Rows N3, N4; Part C items 1–2.

Check (yes/no). Is "T_1" in the sentence on p.396 the fine lattice at spacing one fine site (blocks of M fine sites, radius RM fine sites)?

A4 The region Λ (`LambdaOK2`)

Print. CMP99b p.427 (p427:139): "We assume that $\Lambda \subset \Lambda_k = \Omega_k^{(k)}$, Λ is a union of big blocks, and a distance between Λ and Λ_k^c is bigger than RM." CMP96 p.245, Lemma 2.4 (CMP96_p245:195): "Let a set $\Lambda \subset Z^d$ be a sum of blocks, $\Lambda = B(\Lambda')$."

ZStatementP428.lean:189–195,203–205; ZT4A.lean:57–58,88–90

```
def LambdaOK {Lu Lf : PeriodicLatticeExtent} (ell M R k : ℕ) (S : Tower Lf)
    (Lam : Set (PeriodicLatticeSite Lu)) : Prop :=
  BlockClosed M Lam ∧
  ∃ emb : PeriodicLatticeSite Lu → PeriodicLatticeSite Lf,
    (∀ y μ, (emb y μ).val = (ell ^ k) * (y μ).val) ∧
    (∀ y ∈ Lam, emb y ∈ S.Om k) ∧
    (∀ y ∈ Lam, ∀ y' : PeriodicLatticeSite Lu, emb y' ∉ S.Om k → R * M < pdist y y')
def LamPrime {Lu : PeriodicLatticeExtent} (ell : ℕ) (Lam : Set (PeriodicLatticeSite Lu)) :
    Set (PeriodicLatticeSite Lu) :=
  {y | y ∈ CoarseLattice Lu ell ∧ ∀ x ∈ blockOf y ell, x ∈ Lam}
def LamBlockSum {Lu : PeriodicLatticeExtent} (ell : ℕ) (Lam : Set (PeriodicLatticeSite Lu)) : Prop :=
  ∀ x ∈ Lam, ∃ y ∈ LamPrime ell Lam, x ∈ blockOf y ell
def LambdaOK2 {Lu Lf : PeriodicLatticeExtent} (ell M R k : ℕ) (S : Tower Lf)
    (Lam : Set (PeriodicLatticeSite Lu)) : Prop :=
  LambdaOK ell M R k S Lam ∧ LamBlockSum ell Lam
```

In words. Λ is a set of unit-torus sites, a union of M-blocks; scaling a unit site y to the fine site $\ell^k y$ lands in Ω_k; every unit site y' whose image is not in Ω_k is farther than RM (unit sites) from every site of Λ; and Λ is the union of the ℓ-blocks lying inside it ($\Lambda = B(\Lambda')$). The empty Λ is allowed.

Deviation. Row N2: equal to the print; $\Lambda = \varnothing$ allowed (inequality then trivial, `OneGap.instIneq_empty`). The collar uses the same R as the tower; this is matched by reading, not by a theorem.

Check (yes/no). Is "RM in the unit lattice" the same R as in the tower condition (2.2)?

A5 The class of cubes (IsClassCube)

Print. CMP99b p.396, l.6–10: "For each cube $\square$ of this class there exists a unique index j, $0 \le j \le k$, such that $\square \subset B^j(\Lambda_j) \cup B^{j+1}(\Lambda_{j+1})$, $\square \cap B^j(\Lambda_j) \ne \emptyset$, and $\square$ is a union of several big blocks of the lattice $T_{L^{-j}}$, which implies that its size in the lattice T_η is $O(1)ML^j\eta$. Here $O(1)$ will mean a number ≥ 10." p.408/409 (p408:131, p409:120): "We have assumed that the number $O(1)$ in the condition (3.35) can be taken as equal to 12".

ZStatementP428.lean:131–137,140–142

```
def DescribedAt {L : PeriodicLatticeExtent} (ell M f k : ℕ) (S : Tower L)
    (sites : Set (PeriodicLatticeSite L)) (i : ℕ) : Prop :=
  i ≤ k ∧
  (∃ z : PeriodicLatticeSite L, GridCorner (bigSide M ell i) z ∧
     sites = siteBox z (f * bigSide M ell i)) ∧
  sites ⊆ towerBj S i ∪ towerBj S (i + 1) ∧
  (sites ∩ towerBj S i).Nonempty
def IsClassCube {L : PeriodicLatticeExtent} (ell M f k : ℕ) (S : Tower L)
    (sites : Set (PeriodicLatticeSite L)) (j : ℕ) : Prop :=
  DescribedAt ell M f k S sites j ∧ ∀ i : ℕ, DescribedAt ell M f k S sites i → i = j
```

In words. With a cube factor f: a class cube of index i is the box of side $f \cdot M\ell^i$ fine sites whose corner has coordinates divisible by $M\ell^i$, lying in $(\Omega_i \setminus \Omega_{i+1}) \cup (\Omega_{i+1} \setminus \Omega_{i+2})$, meeting $\Omega_i \setminus \Omega_{i+1}$, with i the only index for which this holds. (Here $B^j(\Lambda_j) = \Omega_j \setminus \Omega_{j+1}$, p.393.) The headline uses the single number $f = 12$.

Deviation. D3: the print attaches $O(1) \ge 10$ to each cube and says "union of several big blocks"; Lean takes one number for all cubes and boxes of f blocks per side. Proved variants with a free factor, per-cube factors, and unions of blocks: Part C, items 3–4. The scale "$T_{L^{-j}}$" is read as blocks of $M\ell^j$ fine sites.

Check (yes/no). Is fixing one cube factor (12, as p.409 says) and taking cubes to be boxes an acceptable reading of the p.396 class for this lemma?

A6 The regularity class (3.35), (3.36) (PerCubeRegular2)

Print. CMP99b p.396 (p396:135–48): "for an arbitrary cube $\square$ of the described above class, and for a configuration U there exists a gauge transformation u on $\square$ such that $U^u = e^{i\eta A}$, and if the index of $\square$ is j, then $|A| < O(1)M\alpha_0(L^j\eta)^{-1}$, $|\nabla^\eta A| < O(1)M\alpha_0(L^j\eta)^{-2}$ on $\square$, where $O(1)M$ is a size of $\square$ in $T_{L^{-j}}$; $|\partial^{\eta *}\partial^\eta A| < O(1)M\alpha_0(L^j\eta)^{-3}$ on $\square$." (3.35), (3.36). Bonds of a set: CMP96 p.224 (CMP96_p224:161): "bonds $b \subset T_\eta$: at least one end-point of b belongs to Ω".

ZStatementP428.lean:145–146; ZR2A.lean:47–66,29–32,40–43

```
def CubeBond {L : PeriodicLatticeExtent} (sites : Set (PeriodicLatticeSite L)) (b : Bond L) : Prop :=
  b.1 ∈ sites ∧ periodicLatticeShift b.1 b.2 ∈ sites
def PerCubeRegular2 {L : PeriodicLatticeExtent} (ell M f k : ℕ) (S : Tower L)
    (α₀ η : ℝ) (U : LinkCfg L) : Prop :=
  ∀ (sites : Set (PeriodicLatticeSite L)) (j : ℕ), IsClassCube ell M f k S sites j →
    ∃ (g : PeriodicLatticeSite L → Matrix.specialUnitaryGroup (Fin 2) ℂ)
      (A : Bond L → Matrix (Fin 2) (Fin 2) ℂ),
      (∀ e : Bond L, CubeBond sites e →
        HT (A e) ∧
        ((periodicGaugeTransform g U e : Matrix.specialUnitaryGroup (Fin 2) ℂ) :
            Matrix (Fin 2) (Fin 2) ℂ)
          = NormedSpace.exp (((η : ℂ) * Complex.I) • A e) ∧
        frobNorm (A e) < (f : ℝ) * M * α₀ * (((ell : ℝ) ^ j * η)^(-1 : ℤ) ^ 1)) ∧
      (∀ (μ ν : Fin 3) (x : PeriodicLatticeSite L),
        CubeBond sites ((x, ν) : Bond L) → CubeBond sites ((periodicLatticeShift x μ, ν) : Bond L) →
        frobNorm ((η^(-1 : ℤ) : ℝ) • (A (periodicLatticeShift x μ, ν) - A (x, ν)))
          < (f : ℝ) * M * α₀ * (((ell : ℝ) ^ j * η)^(-1 : ℤ) ^ 2)) ∧
      (∀ (μ : Fin 3) (x : PeriodicLatticeSite L),
        (∀ ν : Fin 3, ν ≠ μ → CurlReads sites (periodicLatticeShiftBack x ν) ν μ ∧
          CurlReads sites x ν μ) →
        frobNorm (ddA η A x μ)
          < (f : ℝ) * M * α₀ * (((ell : ℝ) ^ j * η)^(-1 : ℤ) ^ 3))
noncomputable def curlA {L : PeriodicLatticeExtent} (η : ℝ) (A : Bond L → Matrix (Fin 2) (Fin 2) ℂ)
    (x : PeriodicLatticeSite L) (a b : Fin 3) : Matrix (Fin 2) (Fin 2) ℂ :=
  (η^(-1 : ℤ) : ℝ) • (A (periodicLatticeShift x a, b) - A (x, b)
    - A (periodicLatticeShift x b, a) + A (x, a))
noncomputable def ddA {L : PeriodicLatticeExtent} (η : ℝ) (A : Bond L → Matrix (Fin 2) (Fin 2) ℂ)
    (x : PeriodicLatticeSite L) (μ : Fin 3) : Matrix (Fin 2) (Fin 2) ℂ :=
  (η^(-1 : ℤ) : ℝ) • ∑ ν ∈ Finset.univ.filter (fun ν : Fin 3 => ν ≠ μ),
    (curlA η A (periodicLatticeShiftBack x ν) ν μ - curlA η A x ν μ)
```

In words. U is regular with parameter α_0 if for every class cube (index j) *there are* a gauge transformation g and a Hermitian traceless field A with $U^g = \exp(i\eta A)$ on the bonds of the cube, $\|A\| < fM\alpha_0(\ell^j\eta)^{-1}$, the forward difference of A (divided by η) is $< fM\alpha_0(\ell^j\eta)^{-2}$ where both bonds are in the cube, and $\partial^*\partial A$, written as the lattice curl-of-curl `ddA`, is $< fM\alpha_0(\ell^j\eta)^{-3}$ where all bonds it reads are in the cube. g and A depend on the cube.

Deviation. N6: "$\exists g, A$ per cube" equals the print. $\sqrt{2}$: norm of A is the full-matrix Frobenius norm, $\sqrt{2}$ times the printed norm (absorbed in α_0). D9: $\partial^*\partial$ replaced by the curl form. Bond convention: `CubeBond` needs *both* endpoints in the cube; CMP96 p.224 says *at least one*. Part C, item 5.

Check (yes/no). (a) Is the existential gauge g chosen per cube, inside the "for every cube"? (b) Is "bonds of the cube = both endpoints in the cube" acceptable for the class conditions?

A7 Coverage and the single hypothesis on U (`ClassCoupled`)

Print. CMP99b p.396, l.after the display (p396:151): "Let us notice that these regularity conditions do not impose any constraints on U outside the domain Ω_0."

Coverage is not printed.

ZR2A.lean:107–109; ZR6A.lean:27–30,33–34,124–125

```
def CoversOm0c {L : PeriodicLatticeExtent} (ell M c k : ℕ) (S : Tower L) : Prop :=
  ∀ x ∈ S.Om 0, ∃ (sites : Set (PeriodicLatticeSite L)) (j : ℕ),
    IsClassCube ell M c k S sites j ∧ x ∈ sites
def CoversPlaqOm0c {L : PeriodicLatticeExtent} (ell M c k : ℕ) (S : Tower L) : Prop :=
  ∀ p : PeriodicLatticePlaquette L, (∀ e ∈ periodicPlaquetteBoundaryEdges p, CubeBond (S.Om 0) e) →
    ∃ (sites : Set (PeriodicLatticeSite L)) (j : ℕ), IsClassCube ell M c k S sites j ∧
      ∀ e ∈ periodicPlaquetteBoundaryEdges p, CubeBond sites e
def CoversOm0p {L : PeriodicLatticeExtent} (ell M c k : ℕ) (S : Tower L) : Prop :=
  CoversOm0c ell M c k S ∧ CoversPlaqOm0c ell M c k S
def ClassCoupled {L : PeriodicLatticeExtent} (ell M c k : ℕ) (S : Tower L) (α₀ η : ℝ) (U : LinkCfg L) : Prop :=
  CoversOm0p ell M c k S ∧ PerCubeRegular2 ell M c k S α₀ η U
```

In words. Every site of Ω_0 lies in a class cube of factor c; every plaquette whose four boundary bonds lie in Ω_0 has its four bonds in one class cube of factor c; and U is regular on all class cubes with the same c. The hypothesis on U in the lemma is this conjunction.

Deviation. D2: an added hypothesis (the print says the conditions constrain U only inside Ω_0; Lean reads: they constrain U throughout Ω_0). It restricts the towers. Without it the encoded statement is false (Part C, item 6).

Check (yes/no). Is "the class cubes cover Ω_0 (sites and plaquettes)" a fair reading of "the conditions are imposed on Ω_0"?

A8 Site operators: Δ'_a, G', R, W, Δ_π

Print. CMP99b p.394 (p394:160-84): "$\Delta'_a = (\Delta_U^\eta + Q'^* a Q') \restriction_{\Omega_0} \dots \langle \lambda, Q'^* a Q' \lambda\rangle = \sum_{j=0}^{k} a_j \sum_{y\in\Lambda_j} (L^j\eta)^{d-2} |(Q'_j(U)\lambda)(y)|^2$ (3.24), $a_{j+1} = a a_j/(a L^{-2} + a_j)$, $a_1 = a_0 = a > 0 \dots$ $\Delta'_a \restriction_{\Omega_0} = \Omega_0 \Delta'_a \Omega_0 \dots$ Its inverse is denoted by G'". (3.21) R is the orthogonal projection onto $\Delta_U N(Q')$ (p.394). (3.119) p.419: $\langle A, \Delta_\pi A\rangle = \langle A - DG'RD^*A, \Delta(A - DG'RD^*A)\rangle$.

R2Ops1.lean:121–125; ZXR2Wire1.lean:20–24; ZR2Ops2.lean:27–30,57–58,63–64; R2Ops1.lean:162–163,170–171; ZR2Ops2.lean:67–68,83–84

```
structure TowerGeom (L : PeriodicLatticeExtent) where
  ell : ℕ
  k : ℕ
  Om0 : Set (PeriodicLatticeSite L)
  Lam : ℕ → Set (PeriodicLatticeSite L)
def geomOf (ell k : ℕ) {Lf : PeriodicLatticeExtent} (S : Tower Lf) : TowerGeom Lf where
  ell := ell
  k := k
  Om0 := S.Om 0
  Lam := fun j => {x | (∀ μ : Fin 3, ell ^ j | (x μ).val) ∧ x ∈ S.Om j ∧ x ∉ S.Om (j + 1)}
noncomputable def Qpa (G : TowerGeom L) (U : Cfg L) (a : ℝ) : SiteL2 L →L[ℝ] SiteL2 L :=
  ∑ j ∈ Finset.range (G.k + 1), ∑ y : PeriodicLatticeSite L,
    if y ∈ G.Lam j then (ajSeq a G.ell j * (G.ell : ℝ) ^ j) •
      ((ContinuousLinearMap.adjoint (Qlev G U j y)).comp (Qlev G U j y)) else 0
noncomputable def DelpA (G : TowerGeom L) (U : Cfg L) (a : ℝ) : SiteL2 L →L[ℝ] SiteL2 L :=
  ((siteMask G.Om0).comp (lapFull U + Qpa G U a)).comp (siteMask G.Om0)
noncomputable def Gp (G : TowerGeom L) (U : Cfg L) (a : ℝ) : SiteL2 L →L[ℝ] SiteL2 L :=
  (Ring.inverse (DelpA G U a + (1 - siteMask G.Om0))).comp (siteMask G.Om0)
noncomputable def lapDir (G : TowerGeom L) (U : Cfg L) : SiteL2 L →L[ℝ] SiteL2 L :=
  ((siteMask G.Om0).comp (lapFull U)).comp (siteMask G.Om0)
noncomputable def Rproj (G : TowerGeom L) (U : Cfg L) : SiteL2 L →L[ℝ] SiteL2 L :=
  (RImage G U).starProjection
noncomputable def Wop (G : TowerGeom L) (U : Cfg L) (a : ℝ) : BondFnL2 L →L[ℝ] BondFnL2 L :=
  1 - (gradL2 U).comp ((Gp G U a).comp ((Rproj G U).comp (gradAdj U)))
noncomputable def DpiFull (G : TowerGeom L) (U : Cfg L) (a : ℝ) : BondFnL2 L →L[ℝ] BondFnL2 L :=
  (ContinuousLinearMap.adjoint (Wop G U a)).comp ((Delta G U).comp (Wop G U a))
```

In words. The tower geometry (Ω_0 and the strata Λ_j at coarse sites) is read off the tower. `Qpa` is the mass term (3.24); `DelpA` is $\Omega_0(\Delta + Q'^* a Q')\Omega_0$; `Gp` is its inverse on $L^2(\Omega_0)$, extended by zero; R is the projection onto $\Delta_U N(Q')$ with Dirichlet Laplacian (`lapDir`); $W = I - DG'RD^*$; $\Delta_\pi = W^*\Delta W$. `Ring.inverse` is Lean's inverse, with value 0 where there is none.

Deviation. D10, D11: the averaging operators $Q_j(U)$ come from the one-step jets of the print's (3.13)–(3.15), per component; convention, bridged but not re-derived. The Lean body is written at $\eta = 1$ ("η-free"; see A12). Not pinned to a printed page: that the weights $L^{j(d-2)}$ versus `ell^j` in `Qpa` equal the print's in $d = 3$.

Check (yes/no). Do `DelpA`, `Gp`, `Wop`, `DpiFull` correspond to (3.24), the inverse G', W of (3.119), with Dirichlet conditions on Ω_0^c?

A9 The operators Δ, J and the plaquette selection

Print. CMP99b p.392 (p392:187-104): $\langle A, \Delta A\rangle = \langle A, D^*DA\rangle + \langle A, \Delta' A\rangle$, $\langle A, \Delta' A\rangle = \sum_{p\subset T_\eta} \eta^d \mathrm{tr}((D^1_U A)(p))^2 \eta^{-2}(\mathrm{Re}\, U(\partial p) - 1) + \dots$ (3.10); $\langle A, J\rangle = \sum_{b\subset T_\eta} \eta^d \mathrm{tr} A(b) J(b)$, $J = D^*\eta^{-2}\mathrm{Im}\,\partial U$ (3.11). p.394 (p394:181-84): "They depend on the configuration U restricted to $\Omega_0 \dots$ All operators we will consider will be defined by using Dirichlet boundary conditions on Ω_0."

The print sums (3.10), (3.11) over all *plaquettes of T_η.*

ZR2Ops2.lean:74–75,79–80; ZXR2Wire2.lean:51–52

```
def plaqIn (S : Set (PeriodicLatticeSite L)) : Set (PeriodicLatticePlaquette L) :=
  {p | ∀ e ∈ periodicPlaquetteBoundaryEdges p, e.1 ∈ S ∧ periodicLatticeShift e.1 e.2 ∈ S}
noncomputable def Delta (G : TowerGeom L) (U : Cfg L) : BondFnL2 L →L[ℝ] BondFnL2 L :=
  AdalyFormal.StatementLaneR6.deltaKLocCLM U (plaqIn G.Om0)
noncomputable def Jf : BondFnL2 (scaleUp P.ell P.k P.Lu) :=
  WithLp.toLp 2 (fun b => -(AdalyFormal.IngredientFamily.JSU2Loc 1 (plaqIn P.G.Om0) P.U b))
```

In words. Δ and J are summed only over plaquettes whose four boundary bonds have both endpoints in Ω_0, so they read U only on Ω_0. (J carries the sign of the chart used for $\mathfrak{su}(2)$.)

Deviation. D1: a chosen reading of p.394, not the printed sums. Two literal alternatives (all plaquettes; a bond mask inside $W^*\Delta W$) are refuted by Lean theorems (Part C, item 7).

Check (yes/no). Is localising Δ and J to plaquettes inside Ω_0 an acceptable meaning of "the operators depend on U restricted to Ω_0"?

A10 G_1, H_1, $\tilde{D}^{(2)}$, Q, Q_1 (the pieces of (3.156))

Print. CMP99b (3.126) p.420 (p420:182): $HB = GQ^*(QGQ^*)^{-1}B$. (3.128) p.421 (p421:130): $\langle A, G_1^{-1}A\rangle = \langle A, \Delta_\pi A\rangle - 2\langle HC_\pi^{(2)}(A), J\rangle + \|RD^*A\|^2 + a\|QA\|^2$; (3.129): $H_1B = G_1Q^*(QG_1Q^*)^{-1}B$. $G = (\Delta_\pi + DRD^* + Q^*aQ)^{-1}$ (3.122) p.420.

R2Algebra.lean:145–150,157–159; ZXR2Wire2.lean:61–62,77–78,86–87,90–91,71–72

```
noncomputable def Ginv : E →L[ℝ] E := D.Dpi + D.Ts.comp D.T + D.Mass
/-- 'G' (3.122)/(3.27). -/
noncomputable def G : E →L[ℝ] E := Ring.inverse D.Ginv
/-- 'H = G Q* (Q G Q*)^-1' (3.126). -/
noncomputable def H : F →L[ℝ] E :=
  (D.G.comp D.Qs).comp (Ring.inverse ((D.Q.comp D.G).comp D.Qs))
structure IsG1inv (G1inv : E →L[ℝ] E) : Prop where
  symm : ∀ x y, ⟪G1inv x, y⟫ = ⟪x, G1inv y⟫
  form : ∀ A, ⟪A, G1inv A⟫ = D.K A
noncomputable def G1inv : ↑(bondSupp P.G) →L[ℝ] ↑(bondSupp P.G) :=
  Classical.choose (P.data.exists_G1inv P.data_laws)
noncomputable def Qin : BondFnL2 (scaleUp P.ell P.k P.Lu) →L[ℝ] BondFnL2 P.Lu :=
  P.tau.comp P.sector.Qfull
noncomputable def H1in : BondFnL2 P.Lu →L[ℝ] BondFnL2 (scaleUp P.ell P.k P.Lu) :=
  (iE P.G).comp ((P.data.H1 P.G1inv).comp (ContinuousLinearMap.adjoint P.tau))
noncomputable def D2in (B : BondFnL2 P.Lu) : BondFnL2 P.Lu :=
  WithLp.toLp 2 (((P.ell : ℝ)^-1) · hLift P.Lam P.Q1 (C2tilde P.ell P.unitBgU (WithLp.ofLp B)))
noncomputable def Q1 : (Bond P.Lu → SuTwo) →₁[ℝ] (CoarseBond P.Lu P.ell → SuTwo) :=
  Q1jet P.ell P.unitBgU
```

In words. $G^{-1} = \Delta_\pi + DRD^* + Q^*aQ$ and G its inverse. G_1^{-1} is *the* symmetric operator whose quadratic form is the right side of (3.128) (the Lean choice is the unique such operator, proved to exist); G_1 its inverse. H_1, $\tilde{D}^{(2)}$ (a quadratic map built from the second-order term of the averaging, divided by ℓ), J and the unit-level Q_1 (the linear part of the one-step averaging, divided by ℓ) are built from the same ingredients; Q is the multi-level constraint operator followed by τ, which reads the top level.

Deviation. D10, D11 (jets, convention). The embedding τ and the placing of B at the top level are not in the print (A12). Not pinned: the identification of $\tilde{D}^{(2)} = hC^{(2)}$ with $h = L^{-1}\iota H_0^{-1}\rho$ (cited in the Lean comment to CMP109 p.267, a paper outside the corpus pages used here).

Check (yes/no). Does the chain $G^{-1} \to G \to G_1^{-1} \to G_1 \to H_1$ with QG_1Q^* inverted follow (3.122), (3.126)–(3.129)?

A11 The subspace, $\tilde{B}$ and the operator C

Print. CMP99b p.428 (p428:135): "This form is considered on the subspace $\{B : B = 0$ on $\Lambda^c,\ B = 0$ on $\bigcup_{y\in\Lambda'} Ax(y),\ Q_1B = 0\}\ldots$ variables B restricted to the set of bonds $\tilde{\Lambda} = \Lambda\backslash(\bigcup_{y\in\Lambda'} Ax(y) \cup \bigcup_{c\in\Lambda'} B(c)\cap c)\ldots$ $B = C\tilde{B}$, where C is a linear operator. It is an identity operator on almost all bonds, except the bonds b_0 for which a value $(C\tilde{B})(b_0)$ is equal to a solution of the equation $(QB)(c) = 0$." CMP96 p.245: "a set of bonds b such that at least one of the end-points b_-, b_+ belongs to Λ".

ZStatementP428.lean:211–213,216–218,230–233; ZR2A.lean:121–123; ZXR2Unit2.lean:98–101,104–107

```
def CRowIn {Lu : PeriodicLatticeExtent} (ell : ℕ) (Lam : Set (PeriodicLatticeSite Lu))
    (c : CoarseBond Lu ell) : Prop :=
  c.1.1 ∈ LamPrime ell Lam ∨ AdalyFormal.Balaban.CoarseCurl.coarseShift c.1.1 ell c.2 ∈ LamPrime ell Lam
def IsDroppedIn {Lu : PeriodicLatticeExtent} (Lam : Set (PeriodicLatticeSite Lu)) (ell : ℕ)
    (b : Bond Lu) : Prop :=
  ∃ c : CoarseBond Lu ell, CRowIn ell Lam c ∧ b = droppedBondLast ell c
def OnLambdaTildeIn {Lu : PeriodicLatticeExtent} (Lam : Set (PeriodicLatticeSite Lu)) (ell : ℕ)
    (B : Bond Lu → SuTwo) : Prop :=
  (∀ y ∈ LamPrime ell Lam, ∀ x ∈ blockOf y ell, ∀ b ∈ staircasePath y x, B b = 0) ∧
  (∀ b : Bond Lu, IsDroppedIn Lam ell b → B b = 0)
def OnLambdaT {Lu : PeriodicLatticeExtent} (Lam : Set (PeriodicLatticeSite Lu))
    (B : Bond Lu → SuTwo) : Prop :=
  ∀ b : Bond Lu, ¬ (b.1 ∈ Lam ∨ periodicLatticeShift b.1 b.2 ∈ Lam) → B b = 0
noncomputable def hLift {ell : ℕ} (Lam : Set (PeriodicLatticeSite L))
    (Q : (Bond L → SuTwo) →₁[ℝ] (CoarseBond L ell → SuTwo)) :
    (CoarseBond L ell → SuTwo) →₁[ℝ] (Bond L → SuTwo) :=
  iotaD ell Lam ∘₁ H0inv Lam Q ∘₁ rhoD ell Lam
noncomputable def Cop {ell : ℕ} (Lam : Set (PeriodicLatticeSite L))
    (Q : (Bond L → SuTwo) →₁[ℝ] (CoarseBond L ell → SuTwo)) :
    (Bond L → SuTwo) →₁[ℝ] (Bond L → SuTwo) :=
  LinearMap.id - hLift Lam Q ∘₁ Q
```

In words. $\tilde{B}$ is an $\mathfrak{su}(2)$-valued field on unit bonds that vanishes on the staircase paths $Ax(y)$ of each $y \in \Lambda'$, on each dropped bond $b_0(c)$ of a coarse bond c having an endpoint in Λ', and on every bond with no endpoint in Λ. $C = \mathrm{id} - hQ_1$ where h solves the constraint on the dropped bonds; it is the identity elsewhere. (The inverse of the constraint matrix H_0 is the junk 0 if H_0 is not bijective.)

Deviation. N1: "touching" (at least one endpoint) for $\tilde{B}$ and c, per CMP96 p.245; this is a larger set of $\tilde{B}$ than "$B = 0$ on Λ^c" with both endpoints (Lean `InstIneq.toBoth`: the touching form implies the both-endpoint form).

Check (yes/no). Is $\tilde{B}$ (vanishing on $Ax(y)$ and on the dropped bonds b_0, with the "at least one endpoint" bond convention) the variable set of p.428, and is $C = \mathrm{id} - hQ_1$ "the operator that solves $(QB)(c) = 0$ on b_0"?

A12 The form, the norm, the inequality

Print. CMP99b (3.156) p.428: $\langle B, (QG_1Q^*)^{-1}B\rangle - a\langle B, B\rangle - 2\langle H_1\tilde{D}^{(2)}(B), J\rangle = \langle B, \Delta_k B\rangle$. Norm: p.390 (p390:129): "a norm $|X|$ of a $N \times N$ matrix means the Hilbert–Schmidt norm: $|X|^2 = \mathrm{tr}X^*X$"; p.392 (p392:161): "the trace is normalized, i.e., $\mathrm{tr}\,1 = 1$".

ZXG3A.lean:50–51,54–55,58–63–66; RegularUPerturbation.lean:355–356

```
noncomputable def firstTerm (B : BondFnL2 P.Lu) : ℝ :=
  P.eta * ⟪ContinuousLinearMap.adjoint P.tau B, Ring.inverse (Nmat P) (ContinuousLinearMap.adjoint P.tau B)⟫
noncomputable def genForm (B : BondFnL2 P.Lu) : ℝ :=
  firstTerm P B - P.a * ⟪B, B⟫ - 2 * ⟪P.H1in (P.D2in B), P.Jf⟫
noncomputable def genC : (Bond P.Lu → SuTwo) →₁[ℝ] (Bond P.Lu → SuTwo) := Cop P.Lam P.Q1
abbrev gparams (ell k : ℕ) (Lu : PeriodicLatticeExtent) (hell : 0 < ell)
    (S : Tower (scaleUp ell k Lu)) (U : LinkCfg (scaleUp ell k Lu)) (a : ℝ)
    (Lam : Set (PeriodicLatticeSite Lu)) : IngrBParams :=
  ⟨ell, k, Lu, hell, S, U, a, Lam⟩
noncomputable def bondEnergy (A : Bond L → SuTwo) : ℝ :=
  ∑ b : Bond L, ‖A b‖ ^ 2
```

In words. `genForm` is the form of (3.156), with B placed at the top level by τ^* and the first term multiplied by η; the bilinear pairing of B with itself is the unit-bond L^2 pairing. `genC` is C. `bondEnergy` $\tilde{B} \mapsto \sum_b \|\tilde{B}(b)\|^2$, with the norm on $\mathfrak{su}(2)$ equal to the printed trace-normalised Hilbert–Schmidt norm (Lean `norm_sq_eq_half_frobSq`). The inverse is Lean's: where QG_1Q^* is not invertible the first term is 0.

Deviation. The factor η and τ^* in the first term are the printed units, not a deviation (d=3): the print pairs fields with weight η^d (p.392 (3.11); p.393) and gives derivatives a factor η^{-1} (p.392 (3.8)); every operator in (3.128) is η^{d-2} times its unit-spacing form, so $\langle B, (QG_1Q^*)^{-1}B\rangle$ in print units equals $\eta\langle\tau^*B, (QG_1Q^*)^{-1}_{\text{unit}}\tau^*B\rangle$, while $a\langle B,B\rangle$ and the J term carry no η (Lean: `firstTerm_eq_print`, `H1_eq`, J scaling and $\langle B,B\rangle = \text{bondEnergy}(B)$). τ^* is zero-extension of B to the top level k, where all weights are 1. Read, not proved: the normalisation of $\tilde{D}^{(2)}$ and of $C^{(2)}$ below level k. The junk value 0 makes the statement stronger than one that presupposes existence of the inverses.

Check (yes/no). Is $\|\tilde{B}\|^2 = \sum_b \|\tilde{B}(b)\|^2$ with trace-normalised norm the norm of the printed lower bound $\gamma_0\|\tilde{B}\|^2$?

A13 The inequality, the inner claim, the constants, the quantifier order

Print. CMP99b p.428 (p428:153): "It is more convenient to work with the operator $\tilde{C}^{(k)}(\Lambda)$, because it is defined by a positive definite operator $C^*\Delta_k C$ with a lower bound $\gamma_0 > 0$ independent of k and U. We have proved it in [4], Lemma 2.4, for operators with $U = 1$. Localizing the operators in Δ_k and using the methods of Sect. B we can prove it for $C^*\Delta_k C$ with an arbitrary configuration U satisfying (3.35), (3.36) with $M\alpha_0$ sufficiently small." R, M: p.408 (p408:131): "We take these numbers as powers of L."

ZT5A.lean:56–61,89–96,170–173,200–201,229–230,291–294

```
def InstIneq (ell k : ℕ) (hell : 0 < ell) (a : ℝ) (Lu : PeriodicLatticeExtent) (γ₀ : ℝ)
    (S : Tower (scaleUp ell k Lu)) (Lam : Set (PeriodicLatticeSite Lu)) (U : LinkCfg (scaleUp ell k Lu)) : Prop :=
  ∀ (Bt : Bond Lu → SuTwo), OnLambdaTildeIn Lam ell Bt → OnLambdaT Lam Bt →
    γ₀ * bondEnergy Bt ≤
      GenP.genForm (gparams ell k Lu hell S U a Lam)
        (WithLp.toLp 2 (GenP.genC (gparams ell k Lu hell S U a Lam) Bt))
def CoercG8At (ell M R k : ℕ) (hell : 0 < ell) (a : ℝ) (Lu : PeriodicLatticeExtent)
    (c : ℕ) (α₀ γ₀ : ℝ) : Prop :=
  TorusOK ell M Lu →
  ∀ (S : Tower (scaleUp ell k Lu)), Admissible ell M R k S →
  ∀ (Lam : Set (PeriodicLatticeSite Lu)), LambdaOK2 ell M R k S Lam →
  ∀ (U : LinkCfg (scaleUp ell k Lu)),
    ClassCoupled ell M c k S α₀ ((ell : ℝ) ^ k)⁻¹ U →
    InstIneq ell k hell a Lu γ₀ S Lam U
def P428Core8At (ell M R : ℕ) (hell : 0 < ell) (a : ℝ) (c : ℕ) : Prop :=
  ∃ a₀ : ℝ, 0 < a₀ ∧ ∃ γ₀ : ℝ, 0 < γ₀ ∧
    ∀ (k : ℕ), 1 ≤ k → ∀ (Lu : PeriodicLatticeExtent) (α₀ : ℝ), 0 < α₀ → (c : ℝ) * M * α₀ ≤ a₀ →
      CoercG8At ell M R k hell a Lu c α₀ γ₀
def SidePow6 (ell M : ℕ) (a : ℝ) : Prop :=
  2 ≤ ell ∧ (∃ m : ℕ, 1 ≤ m ∧ M = ell ^ m) ∧ 0 < a
def P428Genuine8 (ell M R : ℕ) (hell : 0 < ell) (a : ℝ) : Prop :=
  SidePow6 ell M a ∧ P428Core8 ell M R hell a
def P428Target8 : Prop :=
  ∀ (ell : ℕ) (hell : 2 ≤ ell) (a : ℝ), 0 < a →
    ∃ m₀ : ℕ, 1 ≤ m₀ ∧ ∀ m : ℕ, m₀ ≤ m → ∃ R₀ : ℕ, ∀ R : ℕ, R₀ ≤ R →
      P428Genuine8 ell (ell ^ m) R (by omega) a
```

In words. `InstIneq`: for every admissible $\tilde{B}$, $\gamma_0\|\tilde{B}\|^2 \le \text{genForm}(C\tilde{B})$. `CoercG8At`: for tori, towers, regions Λ and configurations U satisfying all the hypotheses above, `InstIneq` holds. `P428Core8At`: there are $a_0, \gamma_0 > 0$ (chosen *before* k) such that for every $k \ge 1$, torus, and $\alpha_0 > 0$ with $cM\alpha_0 \le a_0$, `CoercG8At` holds; the headline has $c = 12$. `P428Target8`: for every $\ell \ge 2$ and $a > 0$ there is m_0 such that for all $m \ge m_0$ there is R_0 such that for all $R \ge R_0$ (with $M = \ell^m$) the above holds.

Deviation. D4: a_0, γ_0 are independent of $k \ge 1$, torus, α_0, tower, Λ, U, $\tilde{B}$ (the print: "independent of k and U"); they may depend on ℓ, a, M, R. D5, D6: all $a > 0$ and all $\ell \ge 2$. D8: $k \ge 1$. N5: $M = \ell^m$. "$M\alpha_0$ sufficiently small" is $12M\alpha_0 \le a_0$ with a_0 existential. "M, R sufficiently large" is $m \ge m_0$, $R \ge R_0$, both existential.

Check (yes/no). Is the prover's order of choice (the constants a_0, γ_0 fixed after ℓ, a, m, R and before k, torus, α_0, tower, Λ, U, $\tilde{B}$) what "independent of k and U" means?

Part B. The whole statement in ordinary notation

Write $M = \ell^m$, $\eta = \ell^{-k}$, $N_\mu = \text{Lu}_\mu + 1$ (unit sites per side), fine torus with $N_\mu\ell^k$ sites per side. **Who chooses:** $\forall$ = the adversary, $\exists$ = the prover; the order is left to right.

$$
\begin{aligned}
&\forall \ell \ge 2\ \forall a > 0\ \ \exists m_0 \ge 1\ \ \forall m \ge m_0\ \ \exists R_0\ \ \forall R \ge R_0 :\\
&\quad \exists a_0 > 0\ \ \exists \gamma_0 > 0\ \ \forall k \ge 1\ \ \forall \text{Lu}\ \ \forall \alpha_0 > 0 \text{ with } 12M\alpha_0 \le a_0 :\\
&\qquad \text{if } \ell \mid N_\mu,\ \ell < N_\mu,\ M \mid N_\mu\ (\mu = 1,2,3),\ \text{then}\\
&\qquad \forall \text{ towers } S = (\Omega_j)_{j\ge 0} \text{ on the fine torus with Admissible}(\ell, M, R, k, S)\\
&\qquad \forall \Lambda \subset \text{unit torus with LambdaOK2}(\ell, M, R, k, S, \Lambda)\\
&\qquad \forall U : \text{fine bonds} \to SU(2) \text{ with ClassCoupled}(\ell, M, 12, k, S, \alpha_0, \eta, U) :\\
&\qquad\quad \forall \tilde{B} : \text{unit bonds} \to \mathfrak{su}(2) \text{ admissible (A11)} :\\
&\qquad\qquad \gamma_0 \sum_b \|\tilde{B}(b)\|^2 \ \le\ \text{genForm}_{(S,U,a,\Lambda)}(C\tilde{B}).
\end{aligned}
$$

where $\text{genForm}(B) = \eta\langle\tau^*B, (QG_1Q^*)^{-1}\tau^*B\rangle - a\langle B,B\rangle - 2\langle H_1\tilde{D}^{(2)}(B), J\rangle$ with all operators built from (S, U, a, Λ) as in A8–A12. $\text{ClassCoupled}(c)$ = [every site and every Ω_0-plaquette lies in a class cube of factor c] $\wedge$ [$\forall$ class cube $\square$ of factor c and index j: $\exists g, A$ with $U^g = e^{i\eta A}$ on $\square$ and (3.35), (3.36) with $cM\alpha_0$]. S ranges over all admissible towers, not over one chosen by the prover; γ_0 does not depend on k, U, S, Λ, $\tilde{B}$, α_0 or the torus. Consequences visible in the text: (i) no hypothesis involves a_0 except $12M\alpha_0 \le a_0$, which is satisfiable for any a_0 by taking α_0 small, so the prover cannot empty the statement by its choice of a_0; (ii) the thresholds m_0, R_0 are existential and unnumbered; (iii) γ_0 may depend on ℓ, a, m, R (the print names neither M nor R).

Part C. Stated readings: what changes if rejected, and what is proved instead

Lean names are in the appendix with file and line. "Proved" means a Lean theorem in the project's rebuilt tree, depending only on the three standard axioms `propext`, `Classical.choice`, `Quot.sound`.

#	reading	if the reader rejects it	proved alternative (Lean)
1	**Reading A of** T_1 (p.396): T_1 is the fine lattice in its own unit; blocks of Ω_0 have M fine sites, radius RM fine sites.	Reading B (T_1 = level-k unit lattice) gives a larger Ω_0 built from blocks of $M\ell^k$ fine sites. The frozen statement itself covers only A.	`target8Layer` (`Target8Layer`) drops the clause (6) and keeps only the strict collar and block closure, so any block-closed Ω_0 with the collar is covered.
2	**Choice of** Ω_0 **(p.396/p.394)**: the layer $\leq RM$ ("e.g."); a thinner layer is permitted by p.394.	A thinner (smaller) block-closed Ω_0 is not covered by the frozen statement. A layer must still satisfy the strict collar R_0M, $R_0 \geq R_1$.	`target8Layer`: `P428Target8` with `Admissible` replaced by `AdmNoBase` (no clause 6; one radius). Class cubes read from the same Ω_0. Two-domain form (operators on a thin Ω_0^{op}, class on a larger $\Omega_0^{cl} \supseteq \Omega_0^{op}$): `target8TwoDom`; for unions, boxes, level-0 caps: `famTwoDom_union`, `famTwoDom_box`, `famTwoDom_cap` . *Not covered*: collars below the sufficient threshold R_1 (existential, no value).
3	**One cube factor** $c = 12$, one factor for the whole class (p.409).	The print attaches $O(1) \geq 10$ to each cube.	`P428Genuine8AllC` (every $c \geq 10$ fixed first; `ZT5A.lean:257`); `P428Target8Total` (`TT_F.lean:29`: constants fixed before $c \geq 10$); `P428Target8FamilyTorus` (`MXQ_Tor.lean:58`): each cube its own factor $f \geq 10$.
4	**Cubes are boxes** of side $fM\ell^j$.	The print says "union of several big blocks".	`P428Target8FamilyTorusU` (`UNI_Tor.lean:73`), with `IsClassUnion`: arbitrary unions of grid cells; implies the box statement. Level-0 cap class (cubes may leave Ω_0; regularity on cube $\cap\,\Omega_0$): `FamTorusCapLayer`, `ClassCoupledCap`. The factor is a label not the size, and the block scale is $M\ell^j$ fine sites (not the literal subscript $T_{L^{-j}}$ with $\eta = \ell^{-k}$).
5	**Bond convention**: class bonds have *both* endpoints in the cube (`CubeBond`); $\tilde{B}$ uses "at least one" (N1). Print: CMP96 p.224 says at least one for domains; CMP102 p.277 states it as a general convention: domains are identified "as sets of bonds with at least one end-point belonging to Ω_j, or sets of plaquettes with at least one corner belonging to Ω_j. This remark applies to other sets also."	For the class: if class bonds must be touching bonds, U is constrained on more bonds, so the hypothesis on U is stronger and the claim is weaker; the frozen "both endpoints" class is the larger class (expected to imply the touching version, not yet a Lean theorem, since the coverage clause also uses `CubeBond`). For the operators: if the domains Ω_j, Λ themselves are read as touching bond and corner plaquette sets, as CMP102 p.277 says, the operators change; see row 7 (plaquette selection).	None proved. `CubeBond` omits a bond incident to it; no theorem converts "both" to "at least one".
6	**Coverage** (A7) is an added hypothesis.	If dropped, the encoded statement is *false*: `P428Target8NC` (coverage dropped) is refuted at $\ell = 3$, $a = \frac{1}{2}$, via an in-class U at which G^{-1} is not invertible (Lean junk value 0). Whether a coverage-free lemma presupposing the printed inverses holds is not decided.	`coverage_necessary` (`P428Target8` $\wedge \neg$`P428Target8NC`), `tip_counterexample`, `not_P428Target8NC`. No variant without coverage.
7	**Plaquette selection** (A9): Δ, J over plaquettes inside Ω_0 (`plaqIn`).	The print sums (3.10), (3.11) over all plaquettes. Literal readings are refuted (the Lean statement fails where G^{-1} is not a unit), but whether the printed claim fails where all inverses exist is not decided.	`P428Target8S selIn selIn = P428Target8` (`rfl`); `not_P428Target8Literal` (all plaquettes); `not_P428Target8LiteralBoth`. No proved variant for the all-plaquette reading.
8	**Bond/norm and normalisations**: Frobenius norm in the class ($\sqrt{2}$, $\sqrt{2}$); η and τ^* in the first term of `genForm`; jets of Q (D10, D11).	Constants change; a_0 absorbs the norm factor. The η-factor and τ^* are the printed units (row A12).	None.
9	**Junk-value inverse** (0 where non-invertible).	A version that presupposes the inverses exist is a different statement; the frozen one is stronger.	`p428Target8Total` proves the form with existence of inverses in the conclusion (`CoercG8AtWD`); `TT_F.lean:29`.

Appendix. Sources (file:line)

Statement: `lean/src/ZT5A.lean` (`P428Target8` at 291; `CoercG8At` 89; `InstIneq` 56; `P428Core8At` 170; `SidePow6` 200; `P428Genuine8` 229). Tower and class: `ZStatementP428.lean`; Λ: `ZT4A.lean`; regularity: `ZR2A.lean`; coverage: `ZR6A.lean`; operators: `ZR2Ops2.lean`, `ZXR2Wire1.lean`, `ZXR2Wire2.lean`, `ZXR2Unit2.lean`, `ZXG3A.lean`; base operators: `R2Ops1.lean`, `R2Algebra.lean`; norm: `RegularUPerturbation.lean:355` (all under `lean/src/`). Printed sources: T. Bałaban, Commun. Math. Phys. 96 (1984) 223–250 and 99 (1985) 389–434, pages as cited in each row. Running `lean/Statement.lean` prints every definition quoted above.